

信号处理原理-07

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“信号处理原理” 是**数学**? **物理**? 还是**计算机**?



物理**现象** $\longleftrightarrow$ 物理**规律** $\longleftrightarrow$ 数学**工具** $\longleftrightarrow$ 计算机**处理**

方法论?

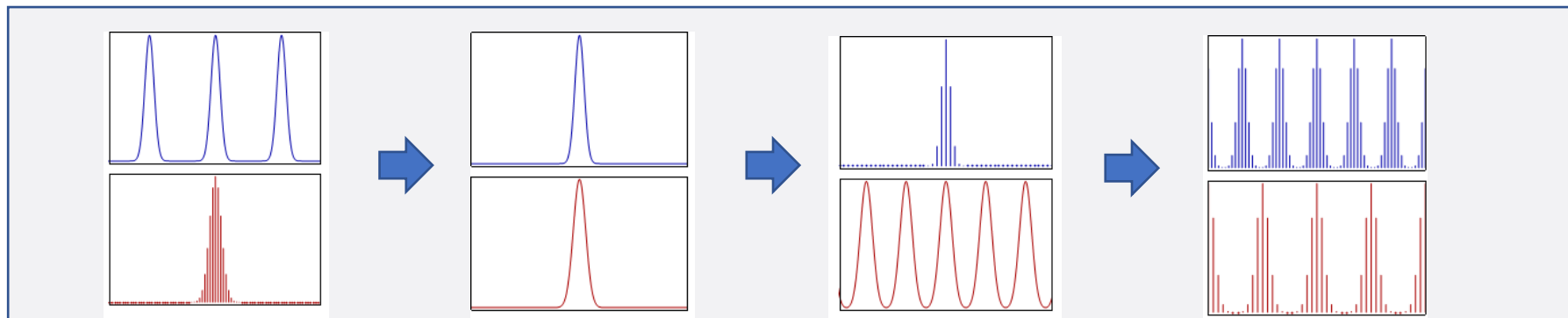
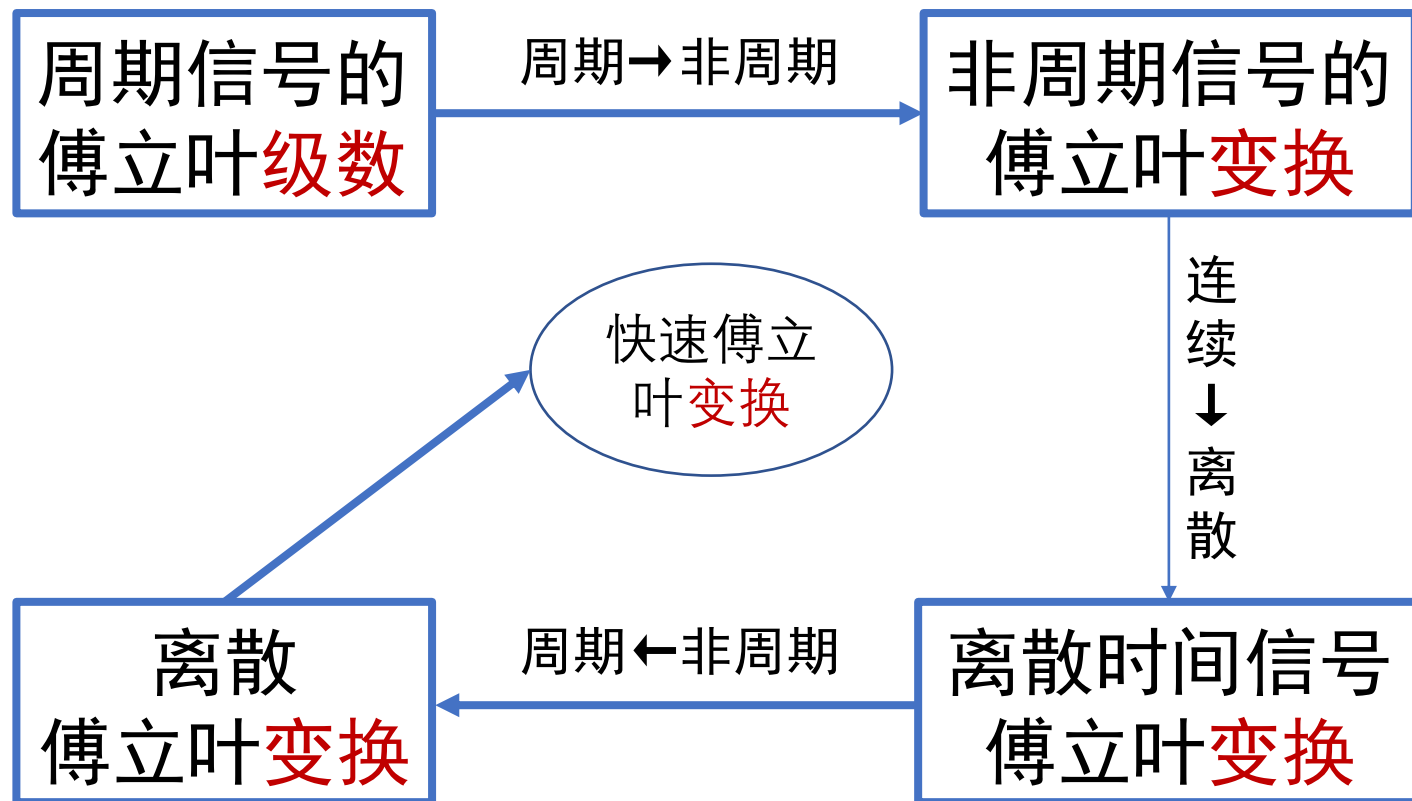
精确or近似?

连续or离散?

- 信号与信号处理的基本概念
- 周期信号的傅里叶级数 FS
- 连续时间傅里叶变换 CTFT
- 离散时间傅里叶变换 DTFT
- 离散傅里叶变换 DFT
- 快速傅里叶变换 FFT
- 离散时间系统分析
- 数字滤波器设计



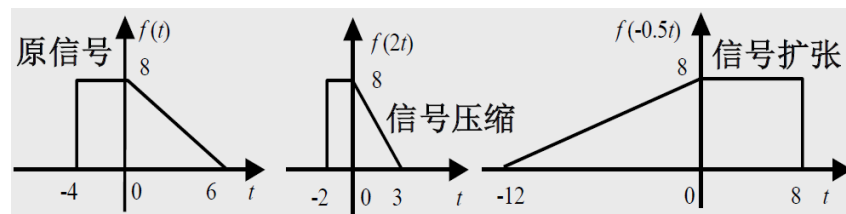
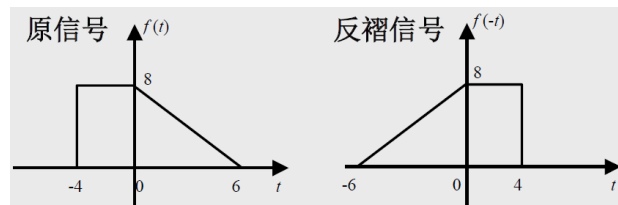
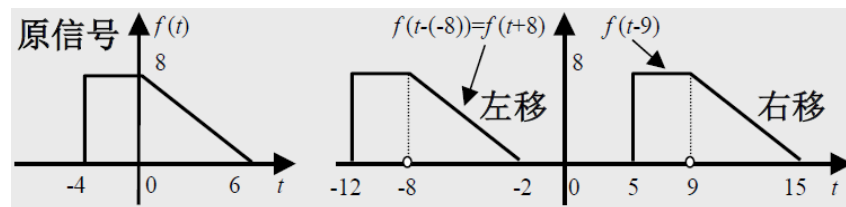
前六周



信号的基本运算

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➤ 关于时间 t 的变换



波形变换

时移运算

$$f(t-t_0)$$

反褶运算

$$f(-t)$$

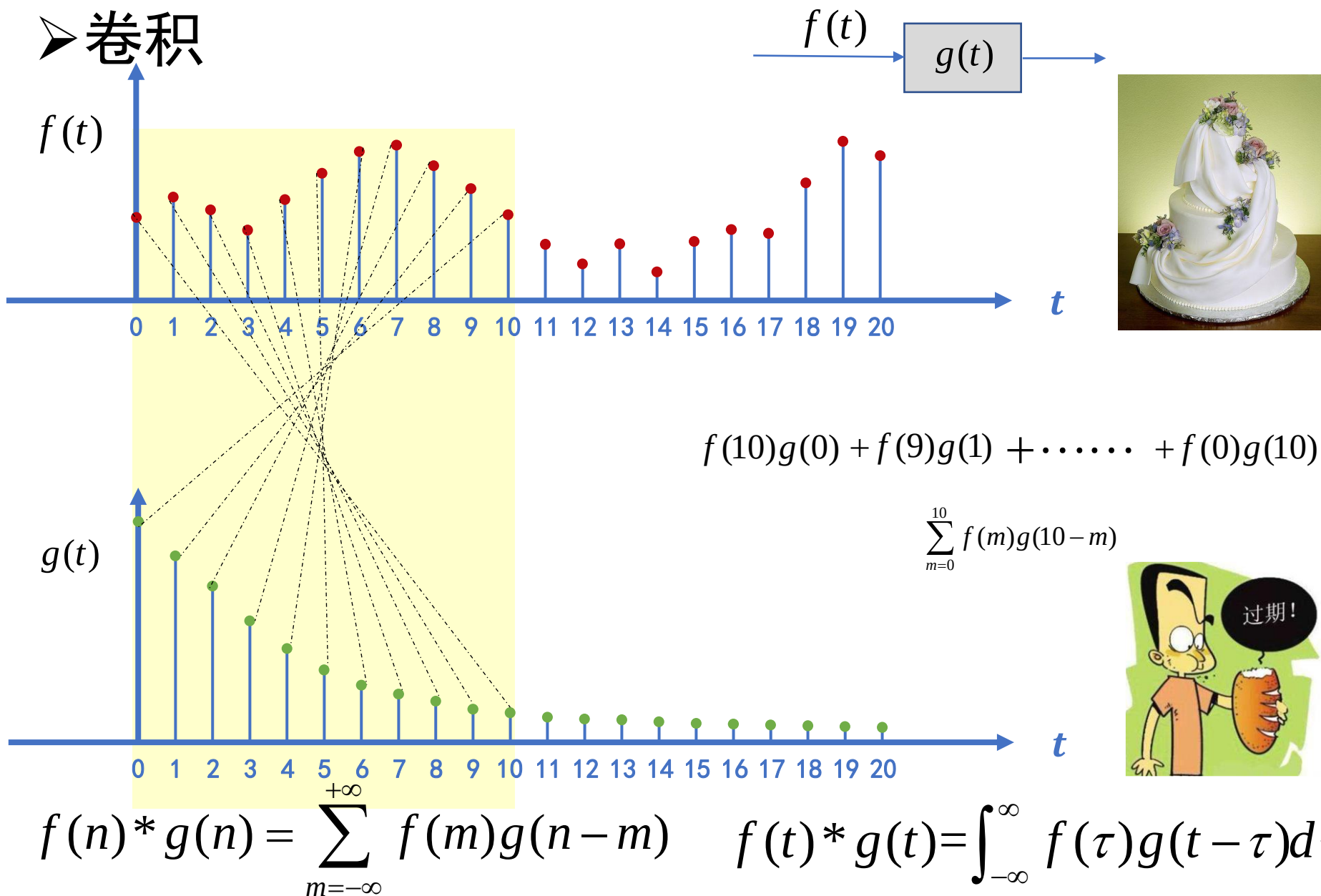
压扩运算

$$f(at)$$

信号的基本运算

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➤ 卷积



➤ 卷积

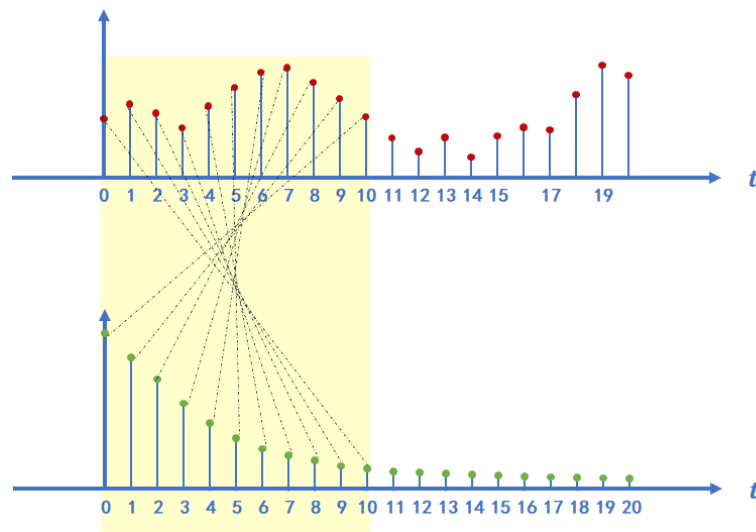
卷积就是一个函数的加权积分

$$f(t) * g(t) = \int_{-\infty}^{\infty} f(\tau) g(t - \tau) d\tau$$

$$t = \tau + (t - \tau)$$

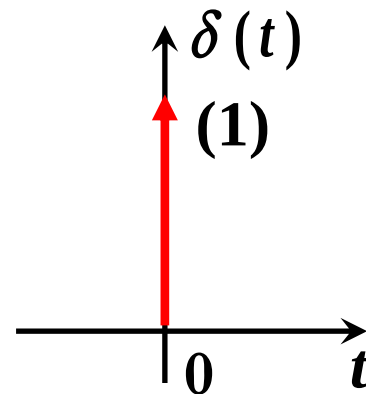
$$n = m + (n - m)$$

$$f(n) * g(n) = \sum_{m=-\infty}^{+\infty} f(m) g(n - m)$$



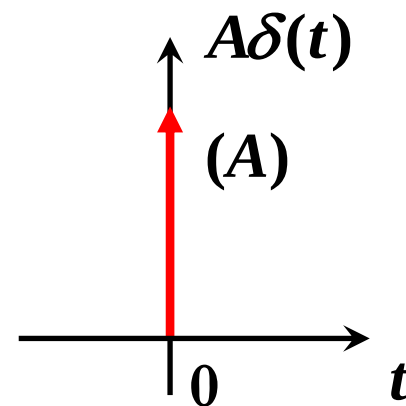
➤ 单位冲激信号

$$\left\{ \begin{array}{ll} \delta(t) = 0 & t \neq 0 \\ \delta(t) = \infty & t = 0 \\ \int_{-\infty}^{\infty} \delta(t) dt = 1 \end{array} \right.$$



➤ 冲激信号

$$\left\{ \begin{array}{ll} A\delta(t) = 0 & t \neq 0 \\ A\delta(t) = \infty & t = 0, \quad A \text{ 为常量} \\ \int_{-\infty}^{\infty} A\delta(t) dt = A \end{array} \right.$$



➤ 筛选性质

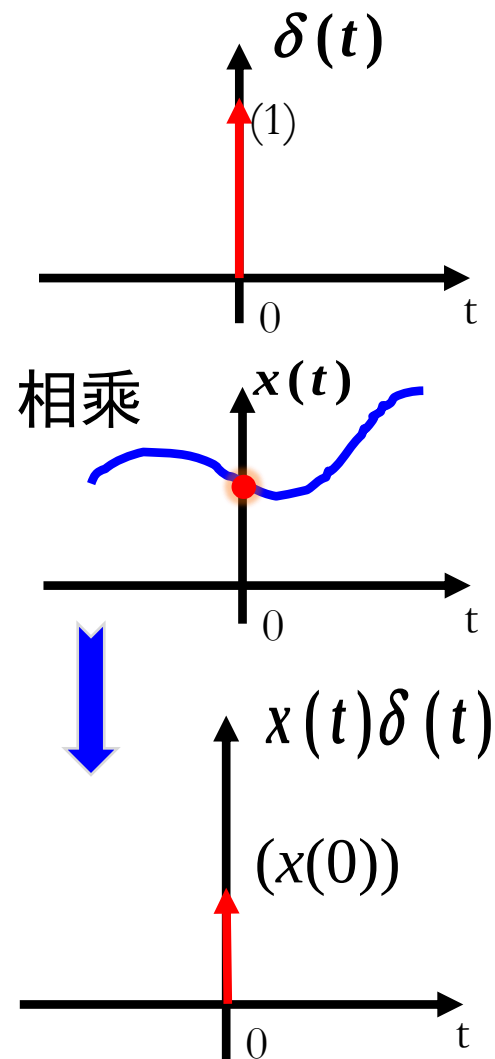
$$x(t)\delta(t) = x(0)\delta(t)$$

$$x(t)\delta(t - t_0) = x(t_0)\delta(t - t_0)$$

➤ 取样性质

$$\int_{-\infty}^{\infty} x(t)\delta(t)dt = x(0)$$

$$\int_{-\infty}^{\infty} x(t)\delta(t - t_0)dt = x(t_0)$$



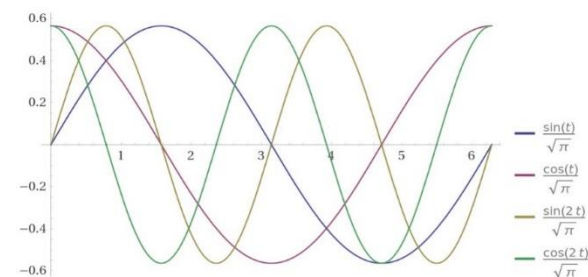
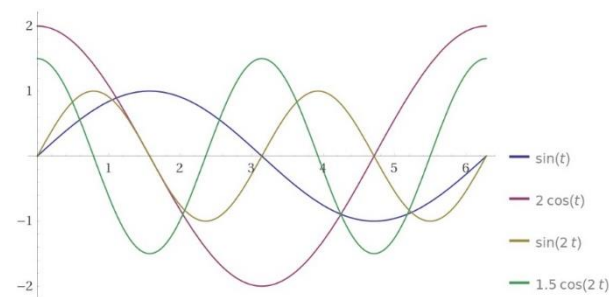
$$f(t) = \sum_{i=1}^{\infty} c_i \varphi_i(t)$$

- 正交函数集

$$\langle \varphi_i, \varphi_j \rangle = \int_{t_1}^{t_2} \varphi_i(t) \varphi_j^*(t) dt = \begin{cases} 0 & i \neq j \\ k_i & i = j \end{cases}$$

- 标准正交函数集

$$\langle \varphi_i, \varphi_j \rangle = \int_{t_1}^{t_2} \varphi_i(t) \varphi_j^*(t) dt = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$$

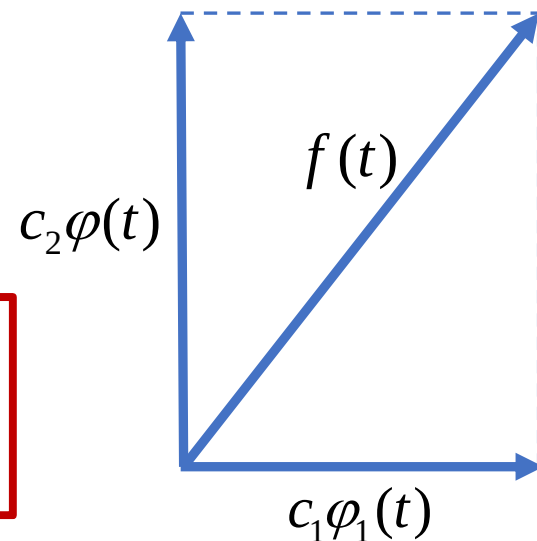


$$f(t) = \sum_{i=1}^{\infty} c_i \varphi_i(t)$$

$$c_j = \frac{\langle f(t), \varphi_j(t) \rangle}{\langle \varphi_j(t), \varphi_j(t) \rangle} = \frac{1}{k_j} \langle f(t), \varphi_j(t) \rangle$$

- 帕斯瓦尔定理

$$\int_{t_1}^{t_2} \|f(t)\|^2 dt = \sum_{i=1}^{\infty} \|c_i\|^2 k_i$$



周期信号的FT

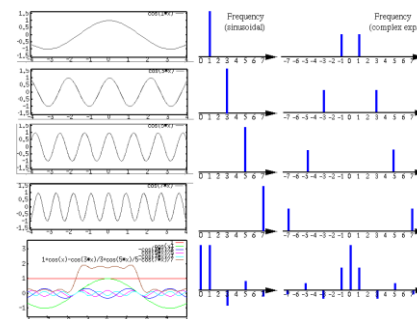
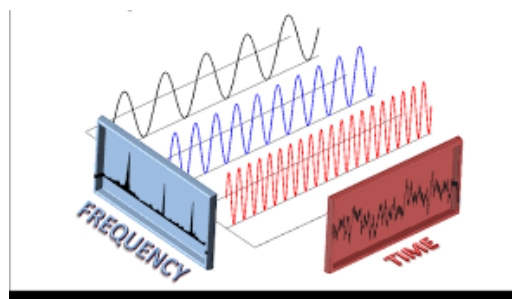
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$$f(t) = \sum_{n=-\infty}^{\infty} c_n \varphi_n(t)$$

$$T_1 = 2\pi / \omega_1$$

三角函数集: $\{1, \cos n\omega_1 t, \sin n\omega_1 t\} \quad n \in \mathbb{N}$

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_1 t + b_n \sin n\omega_1 t)$$



$$a_0 = \frac{1}{T} \int_{t_0}^{t_0+T} f(t) dt$$

$$a_n = \frac{2}{T} \int_{t_0}^{t_0+T} f(t) \cos n\omega_1 t dt$$

$$b_n = \frac{2}{T} \int_{t_0}^{t_0+T} f(t) \sin n\omega_1 t dt$$

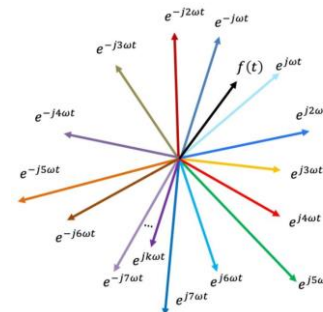
$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_1 t + b_n \sin n\omega_1 t)$$



$$e^{j\omega t} = \cos \omega t + j \sin \omega t$$

$$\cos n\omega_i t = (e^{jn\omega_i t} + e^{-jn\omega_i t}) / 2$$

$$\sin n\omega_i t = (e^{jn\omega_i t} - e^{-jn\omega_i t}) / (2j)$$



$$f(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_1 t}$$

$$F_n = \frac{1}{T_1} \int_{t_0}^{t_0+T_1} f(t) e^{-jn\omega_1 t} dt$$

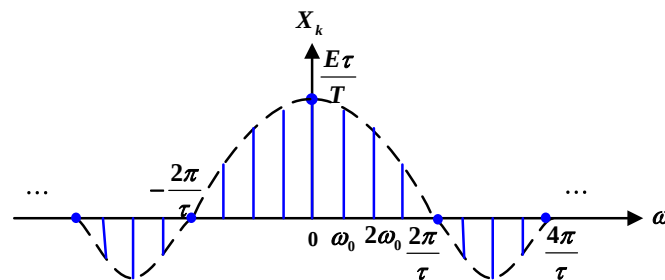
► 帕斯瓦尔定理

$$\int_{t_1}^{t_2} \|f(t)\|^2 dt = \sum_{i=1}^{\infty} \|c_i\|^2 k_i$$

$$f(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_1 t}$$

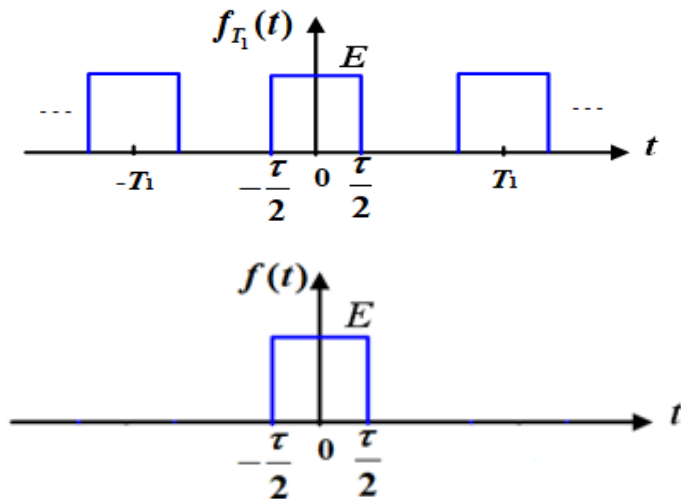
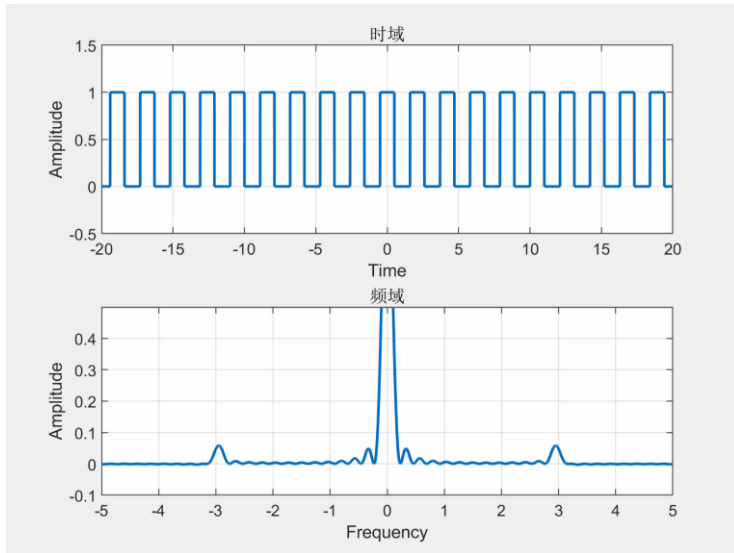
$$F_n = \frac{1}{T_1} \int_{t_0}^{t_0+T_1} f(t) e^{-jn\omega_1 t} dt$$

$$P = \frac{1}{T} \int_{-T/2}^{T/2} |f(t)|^2 dt = \sum_{n=-\infty}^{\infty} |F_n|^2$$



非周期信号的FT

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$$f_{T_1}(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_1 t}$$

$\Downarrow$
 $T_1 \rightarrow \infty$

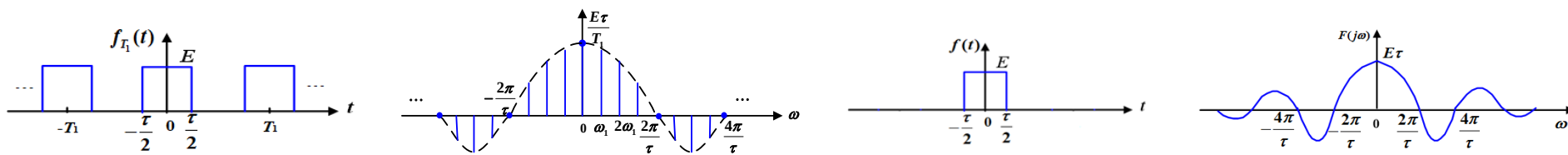
$$F_n = \frac{1}{T_1} \int_{-\frac{T_1}{2}}^{\frac{T_1}{2}} f(t) e^{-jn\omega_1 t} dt \rightarrow 0$$

$$F(\omega) = \lim_{T_1 \rightarrow \infty} T_1 F_n = \lim_{\Delta\omega \rightarrow 0} \frac{2\pi}{\Delta\omega} F_n$$

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(j\omega) e^{j\omega t} d\omega$$

$F(\omega)$ 与 F_n (周期信号频谱) 的区别:



$$F_n = \frac{1}{T_1} \int_{t_0}^{t_0+T_1} f_{T_1}(t) e^{-jn\omega_1 t} dt$$

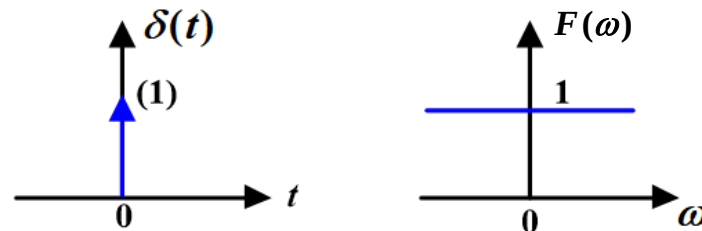
$$f_{T_1}(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_1 t}, \omega_1 = \frac{2\pi}{T_1}$$

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

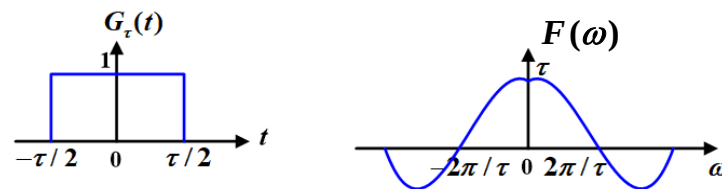
$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega$$

	FS	FT
被分析对象	周期信号	非周期信号
频率定义域	离散频率, 谐波频率处	连续频率, 整个频率轴
函数值意义	频率分量的数值	频率分量的密度值

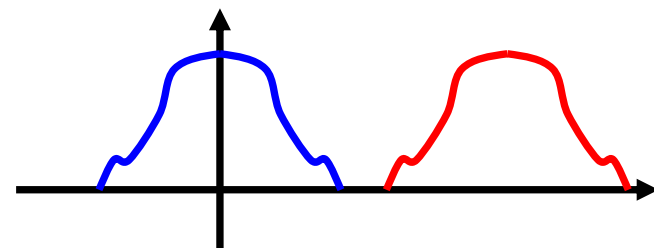
$$\delta(t) \leftrightarrow 1$$



$$G_{\tau}(t) \leftrightarrow \tau \text{Sa}\left(\frac{\omega\tau}{2}\right)$$



$$\mathcal{F}[f(t)e^{j\omega_0 t}] = F(\omega - \omega_0)$$

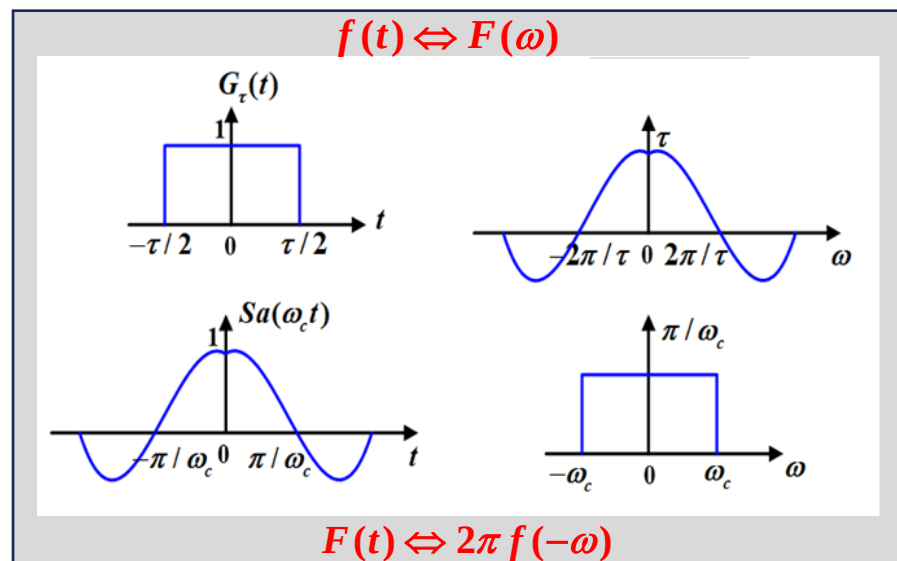
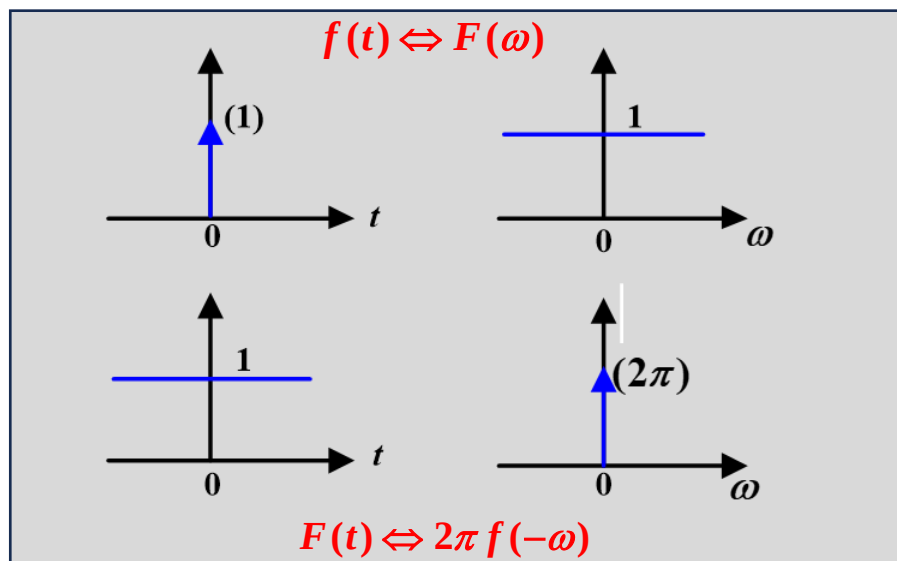


$$\mathcal{F}[f_1(t) * f_2(t)] = \mathcal{F}[f_1(t)] \cdot \mathcal{F}[f_2(t)] \quad \mathcal{F}[f_1(t) \cdot f_2(t)] = \frac{1}{2\pi} \mathcal{F}[f_1(t)] * \mathcal{F}[f_2(t)]$$

非周期信号的FT

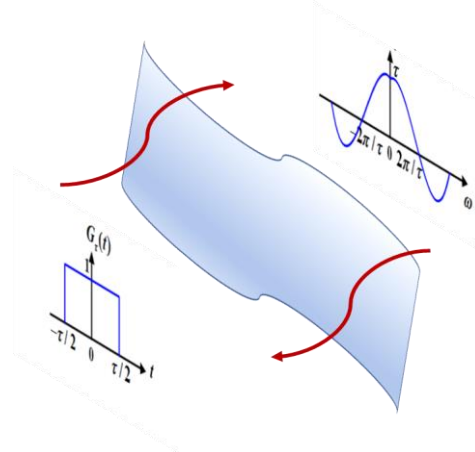
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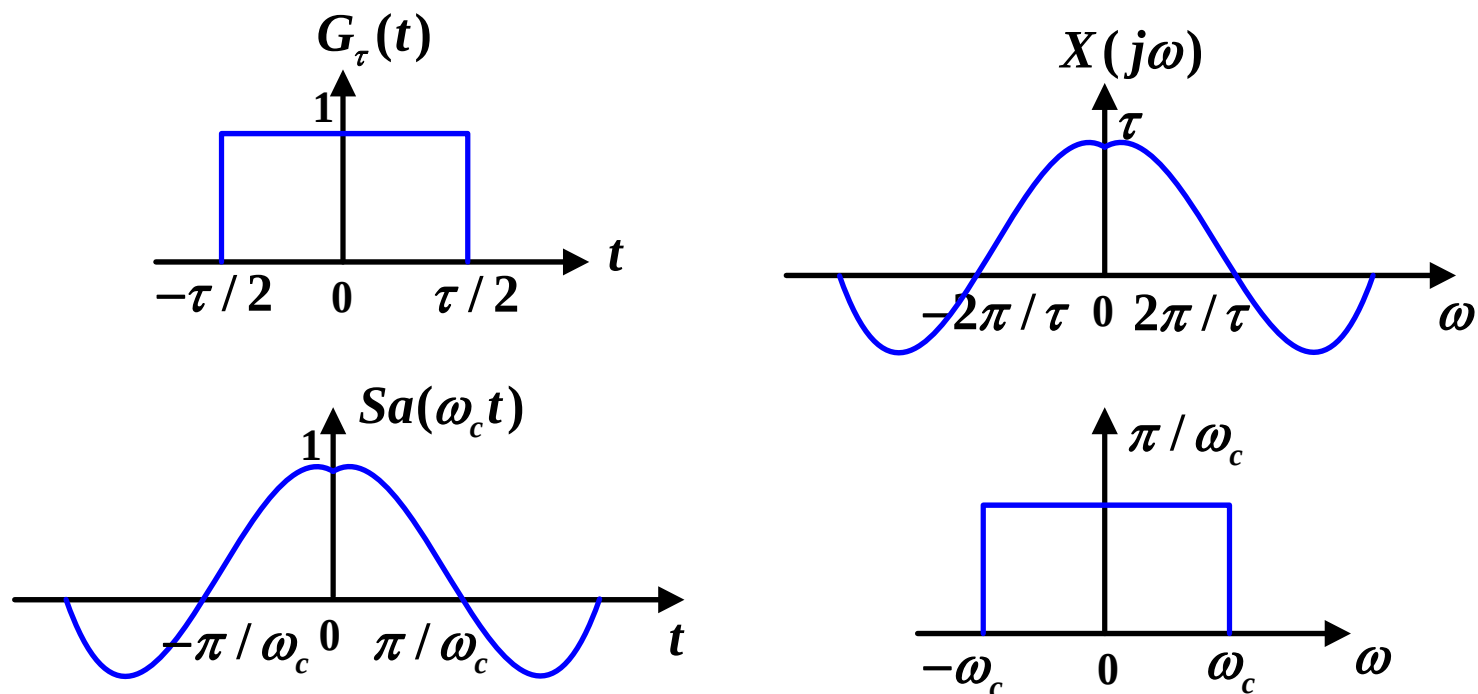
➤ 对偶性质



$$\begin{aligned}
 f(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega \\
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} (F^*(\omega) e^{-j\omega t} d\omega)^* \\
 &= \frac{1}{2\pi} \left(\int_{-\infty}^{\infty} (F^*(\omega) e^{-j\omega t} d\omega) \right)^* = \frac{1}{2\pi} \left\{ \mathcal{F}_{\omega} [F^*(\omega)] \right\}^*
 \end{aligned}$$

两次傅里叶
变换





什么函数（信号）的傅里叶变换是他自身形式？

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega \quad F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

非周期信号的FT

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$$f(t) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{t^2}{2\sigma^2}\right)$$

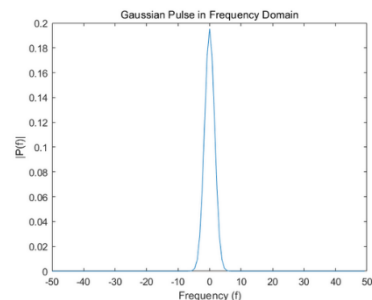
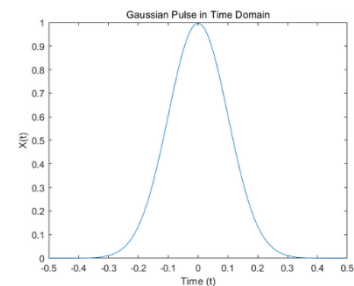
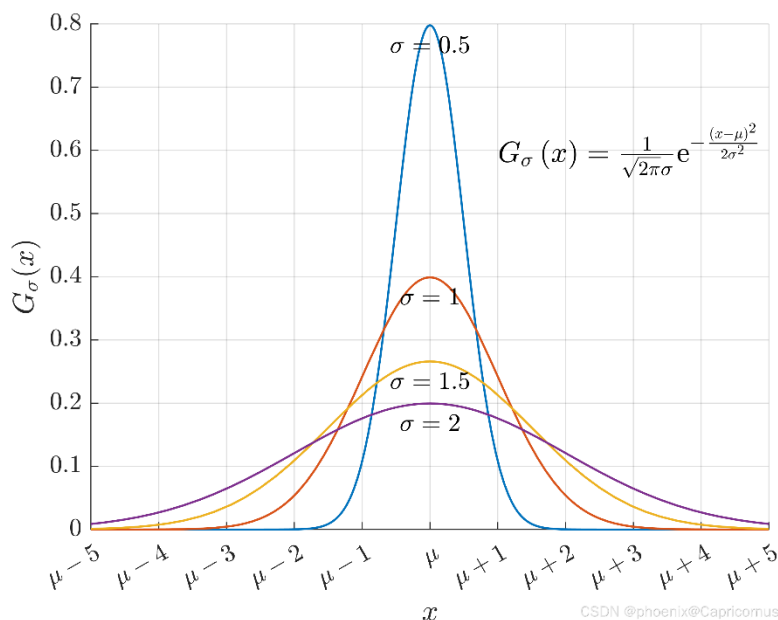
$$F(\omega) = \exp(-2\pi^2\sigma^2\omega^2)$$

$$\sigma = \frac{1}{\sqrt{2\pi}}$$

$$f(t) = \exp(-\pi t^2)$$

$$F(\omega) = \exp(-\pi\omega^2)$$

$$\sigma_t \sigma_\omega = \frac{1}{2\pi}$$



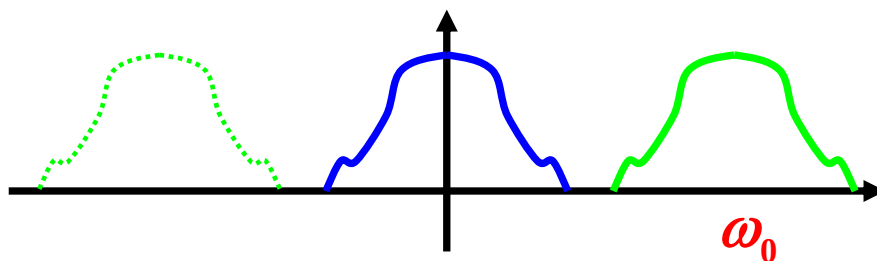
➤ FT的性质：频移特性

若 $f(t) \leftrightarrow F(\omega)$, 则

$$\mathcal{F}[f(t)e^{j\omega_0 t}] = F(\omega - \omega_0) \quad \omega_0 \text{ 为任意实数}$$

证明：

$$\begin{aligned} f(t)e^{j\omega_0 t} &\leftrightarrow \int_{-\infty}^{\infty} [f(t)e^{j\omega_0 t}] e^{-j\omega t} dt \\ &= \int_{-\infty}^{\infty} f(t) e^{-j(\omega - \omega_0)t} dt \\ &= F(\omega - \omega_0) \end{aligned}$$



➤ FT的性质：卷积特性

- 时域卷积定理：

$$\mathcal{F}[f_1(t) * f_2(t)] = \mathcal{F}[f_1(t)] \cdot \mathcal{F}[f_2(t)]$$

- 频域卷积定理：

$$\mathcal{F}[f_1(t) \cdot f_2(t)] = \frac{1}{2\pi} \mathcal{F}[f_1(t)] * \mathcal{F}[f_2(t)]$$

非周期信号的FT

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➤ FT的性质：帕斯瓦尔定理

$$\int_{t_1}^{t_2} \|f(t)\|^2 dt = \sum_{i=1}^{\infty} \|c_i\|^2 k_i$$

$$f(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_1 t}$$

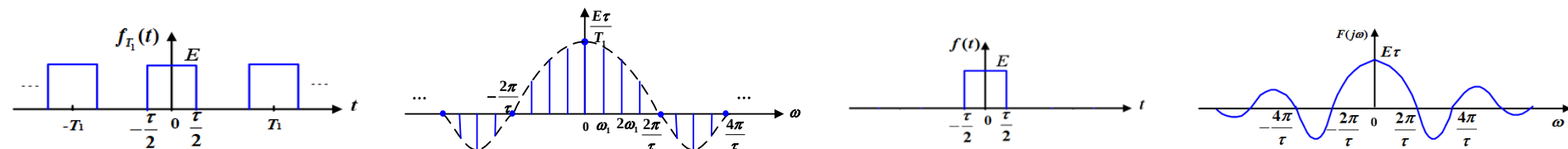
$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega$$

$$F_n = \frac{1}{T_1} \int_{t_0}^{t_0+T_1} f(t) e^{-jn\omega_1 t} dt$$

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

$$\int_{t_0}^{t_0+T_1} \|f(t)\|^2 dt = T_1 \sum_{i=-\infty}^{\infty} \|F_n\|^2$$

$$\int_{-\infty}^{\infty} \|f(t)\|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} \|F(\omega)\|^2 d\omega$$



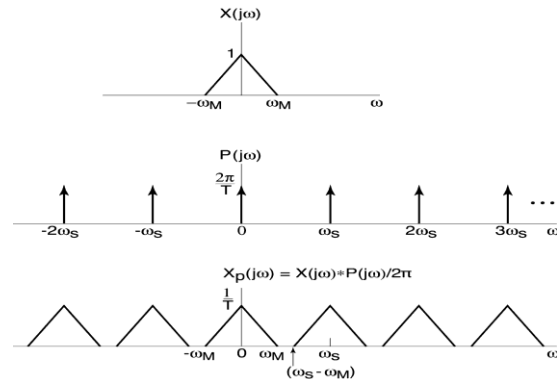
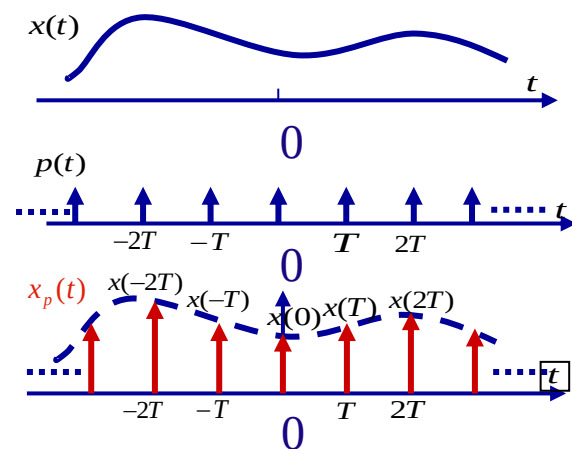
信号的采样

采样信号的时域表示

$$x_p(t) = x(t)p(t) = \sum_{n=-\infty}^{+\infty} x(nT_s)\delta(t - nT_s)$$

采样信号的频域表示

$$X_p(\omega) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} X(\omega - n\omega_s)$$

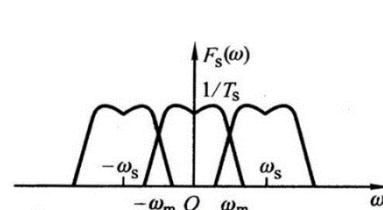
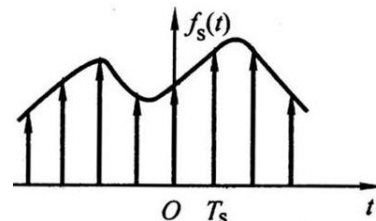
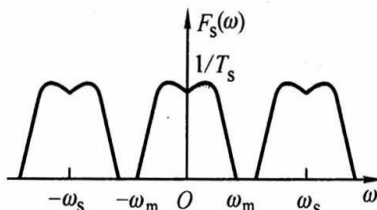
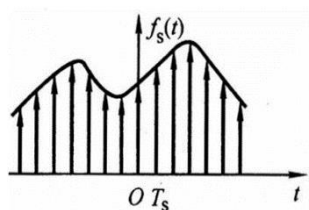


在时域对连续时间信号进行理想采样，就相当于在频域将连续时间信号的频谱以 ω_s 为周期进行延拓。

➤ 信号的采样

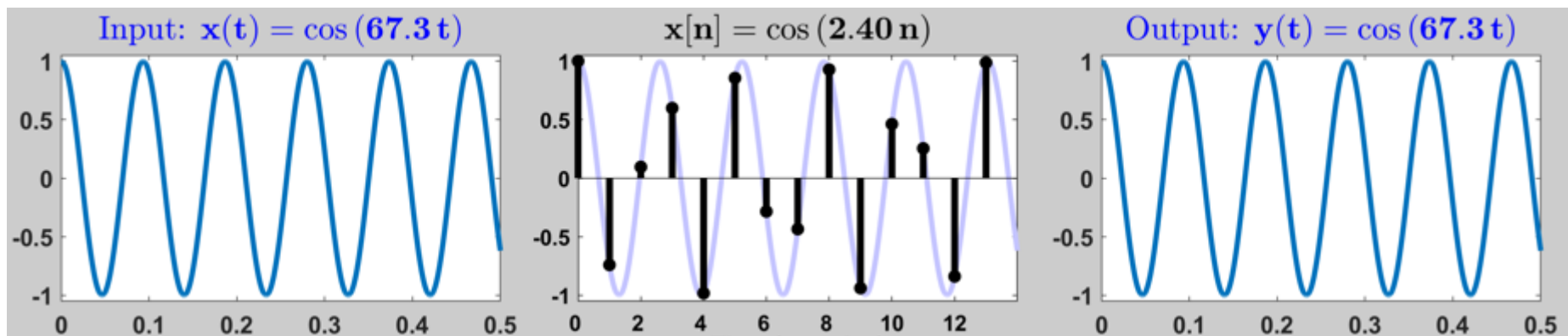
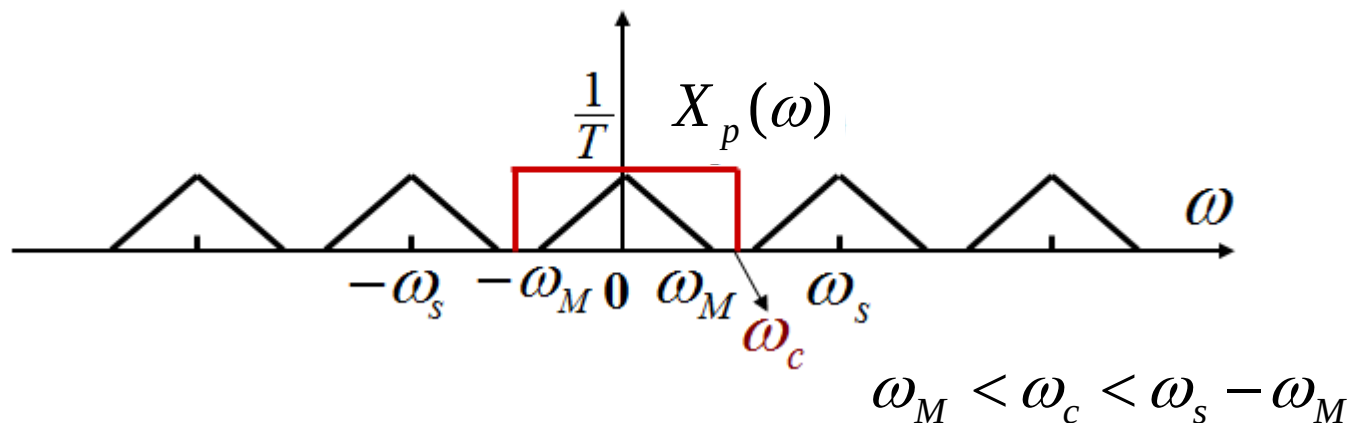
要想使采样后的信号样本能完全代表原来的信号，就意味着要能够从 $X_p(\omega)$ 中不失真地分离出 $X(\omega)$ 。这就要求在周期性延拓时不能发生频谱的混叠。为此必须要求：

- $x(t)$ 必须是带限的，最高频率分量为 ω_M
- 采样频率必须满足 $\omega_s \geq 2\omega_M$



➤ 采样信号的重构

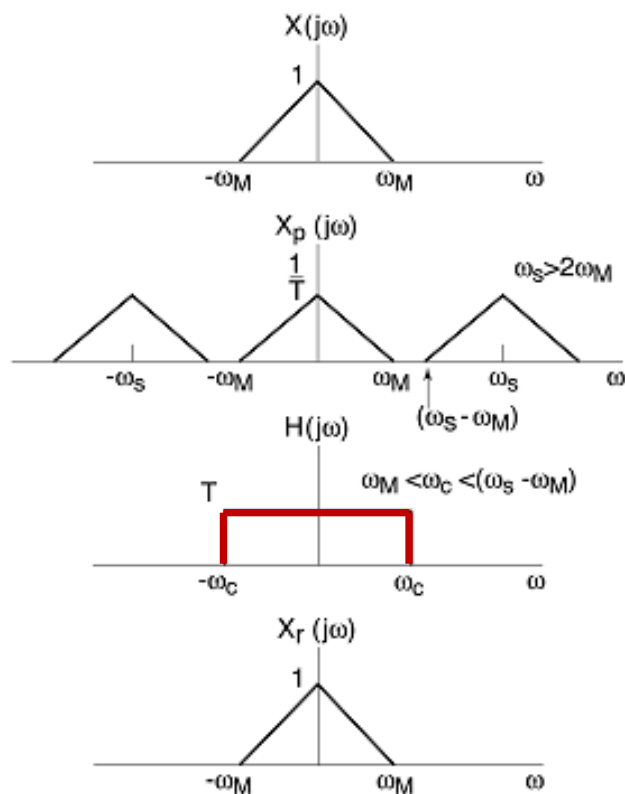
频域 (分析)



时域 (实现)

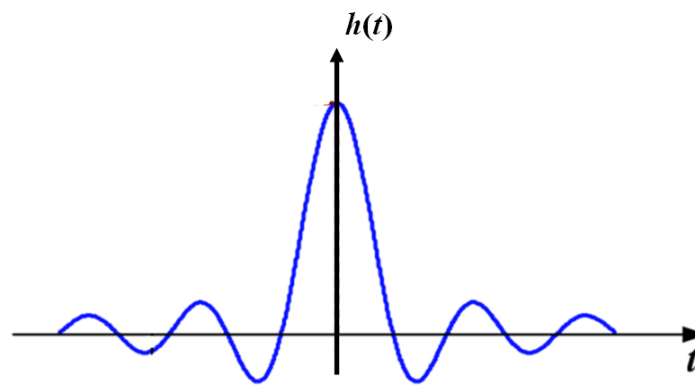
滤波器

➤ 采样信号的重构



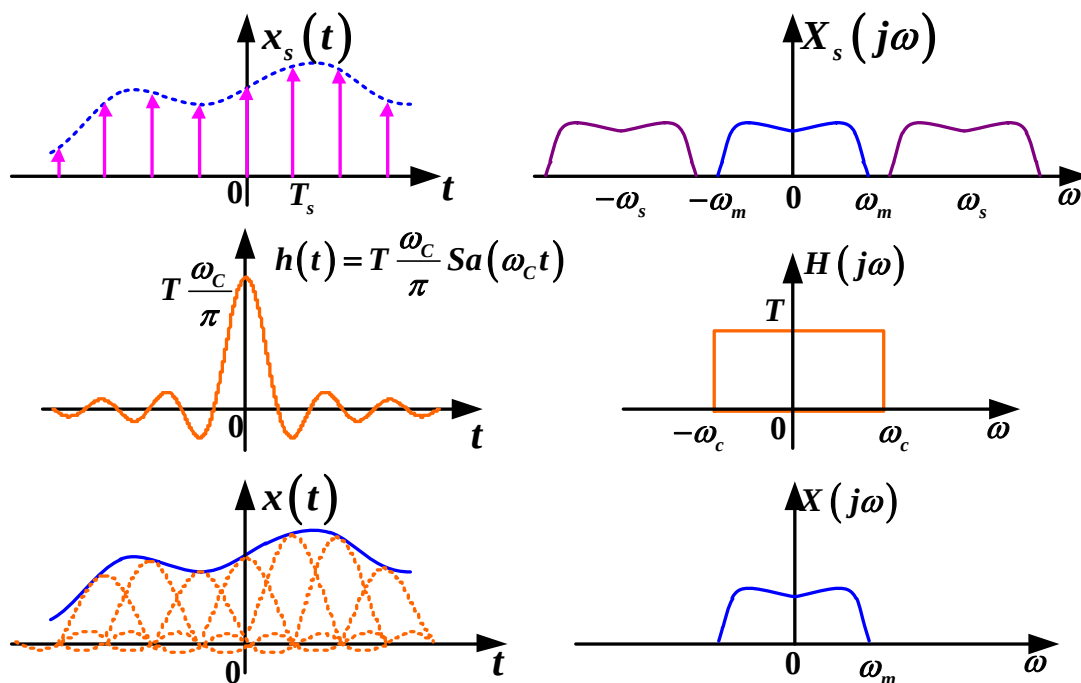
以 ω_c 为截止频率的理想滤波器

$$|H(\omega)| = \begin{cases} T_s & |\omega| \leq \omega_c \\ 0 & |\omega| > \omega_c \end{cases}$$



$$h(t) = T_s \frac{\omega_c}{\pi} \text{Sa}(\omega_c t)$$

➤ 采样信号的重构



$$x_p(t) = \sum_{n=-\infty}^{\infty} x(nT_s) \delta(t - nT_s)$$

$$h(t) = T_s \frac{\omega_c}{\pi} \text{Sa}(\omega_c t)$$

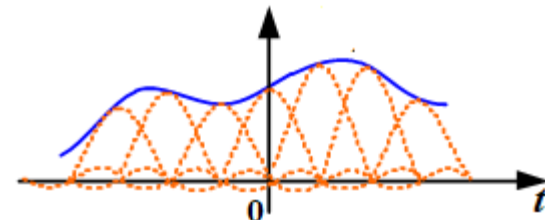
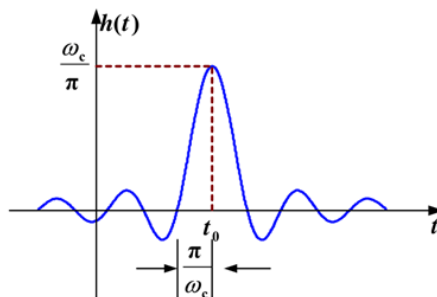
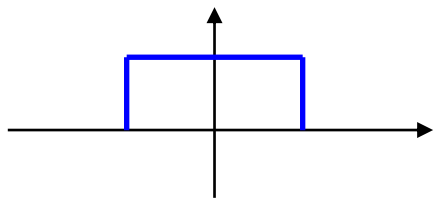
$$\hat{x}(t) = x_p(t) * h(t) = \sum_{n=-\infty}^{\infty} x(nT_s) \delta(t - nT_s) * h(t)$$

$$\hat{x}(t) = \frac{\omega_c T_s}{\pi} \sum_{n=-\infty}^{+\infty} x(nT_s) \text{Sa}(\omega_c (t - nT_s))$$

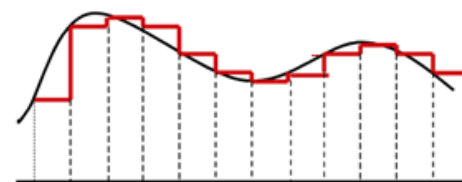
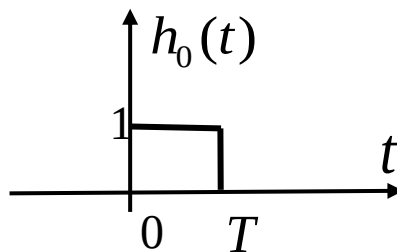
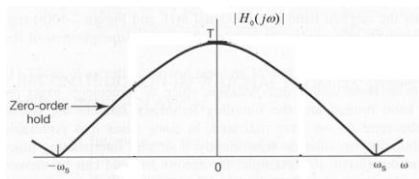
用滤波器函数对信号采样值进行内插来重建连续信号，相当于滤波器的冲击响应与信号为权重的脉冲串的卷积。

➤ 采样信号的重构

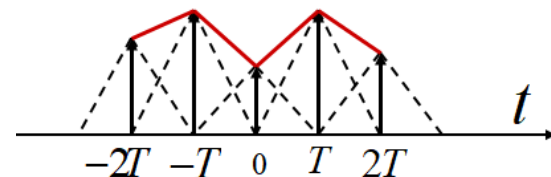
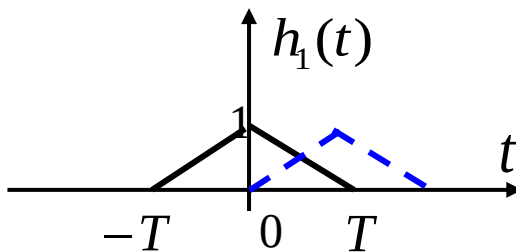
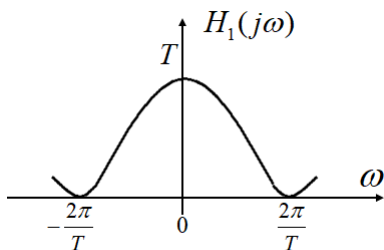
理想

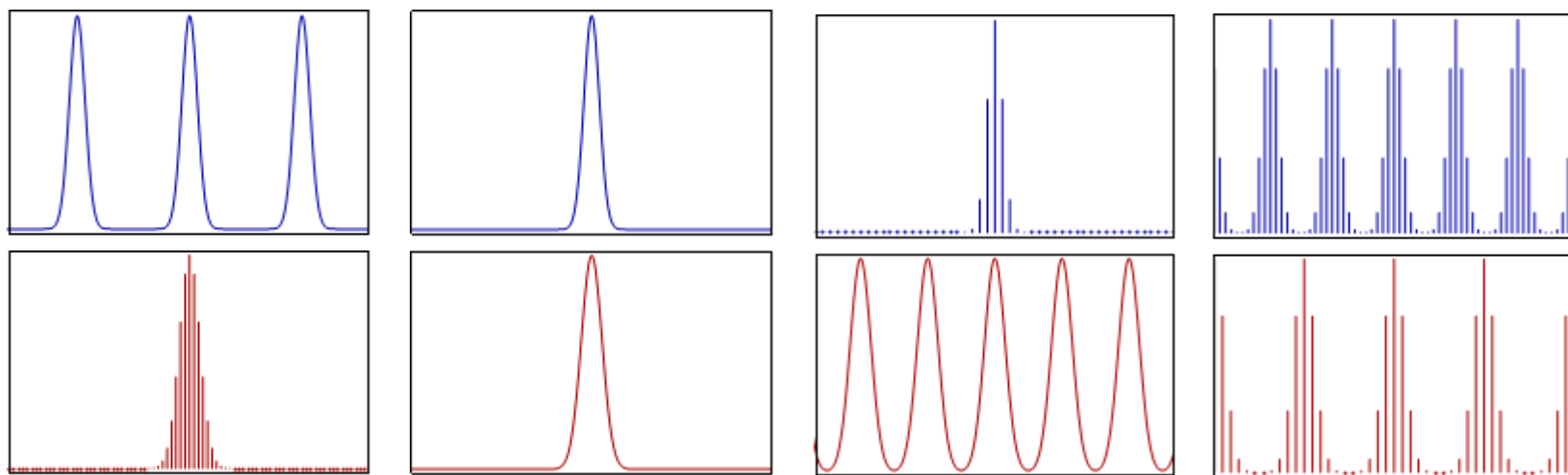


零阶



一阶





$$f(t) = \sum_{i=1}^{\infty} c_i \varphi_i(t)$$

$$\int_{t_1}^{t_2} \|f(t)\|^2 dt = \sum_{i=1}^{\infty} \|c_i\|^2 k_i$$

欧拉公式

傅里叶级数
(三角形式)

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_1 t + b_n \sin n\omega_1 t)$$

傅里叶级数
(复指数形式)

$$f(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_1 t}$$

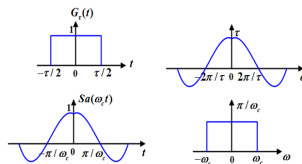
$$F_n = \frac{1}{T_1} \int_{t_0}^{t_0+T_1} f(t) e^{-jn\omega_1 t} dt$$

周期无限
频率密度

傅里叶变换

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega$$

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$



脉冲串
(时域相乘, 频域卷积)

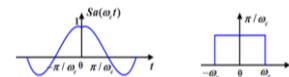
数字频率

信号采样

$$x_p(t) = \sum_{n=-\infty}^{+\infty} x(t) \delta(t - nT_s)$$

$$X_p(\omega) = \sum_{n=-\infty}^{+\infty} X(\omega - n\omega_s)$$

$$\hat{x}(t) = \sum_{n=-\infty}^{\infty} \frac{T_s \omega_s}{\pi} x(nT_s) \text{Sa}[\omega_s(t - nT_s)]$$



离散时间
傅里叶变换

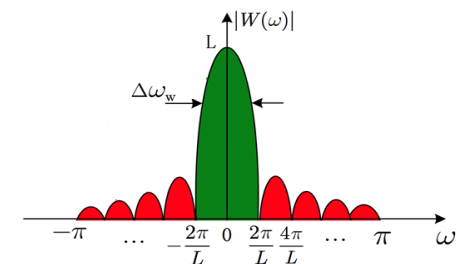
$$X(\Omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\Omega n}$$

$$x(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\Omega) e^{j\Omega n} d\Omega$$

$$\Omega = \omega T_s$$

有限长

加窗



时间：下周课程

地点：上课教室

要求：开卷，可携带教材、纸质ppt，参考资料，计算器

不能使用：计算机，通讯设备

成绩权重：算一次作业